Predictive Cold Chain Disruption Modeling for Pharmaceutical Logistics: An Exploratory Data Analysis Report

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***Abstract*—Pharmaceutical cold-chain logistics is a temperature-sensitive task, where a deviation from optimal conditions may render a vaccine or a sensitive pharmaceutical product ineffective or even dangerous for use. The industry estimates that almost 50% of all global vaccine waste and more than $35 billion USD in pharmaceutical losses per year are caused by temperature control failures. A substantial proportion of these are silent failures that could not be detected by traditional alert systems. In this paper, we demonstrate the results of an exploratory data analysis applied to the Cold Chain Shipment Silent Failure Dataset in order to engineer features for a predictive model of silent failures.**

The dataset contains 8,000 shipment records with 23 different attributes related to sensors, shipment packaging, and routing, along with a binary label indicating whether a silent failure occurred. After conducting necessary preprocessing steps—including type validation, treatment of missing values, univariate, bivariate, and multivariate profiling, and outlier detection—we find that while most attributes are not predictive of failures, time, temperature range, temperature variance, and humidity range are positively correlated with failures. Univariate analysis demonstrates that categorical features such as carrier or zone have little effect on the distribution of failures, while higher values of temperature range, temperature variance, and duration of a shipment are associated with higher rates of failure. Additionally, an engineered feature that estimates the intensity of temperature excursion, calculated as a weighted sum of temperature range and the number of hours since the last sensor reading, encapsulates the concept of a silent failure better than any individual attribute. Its value is positively correlated with the target. This analysis lays the groundwork for an explainable ML/DL-based predictive model for tracking silent failures in time that would be built upon the anomaly detection methodology pioneered by Xie et al.

I. Introduction

Cold-chain logistics for pharmaceuticals is an unusual domain; even a brief exposure to non-regulated temperatures could render a potentially life-saving drug or vaccine ineffective or even dangerous. Industry and public health statistics show that the scale of the problem is deeply concerning: approximately 50% of all vaccines produced are wasted every year, with a considerable proportion tracing their origins to temperature mishaps during storage or transportation [1]. Temperature-related losses are estimated to cost the pharmaceutical industry more than $35 billion USD per year [2]. Compromised temperature-regulated pharmaceuticals not only entail direct economic losses but also have ramifications for the healthcare ecosystem, such as damaging the trust of the general public in the healthcare system and/or legal issues for manufacturers and logistics providers.

Conventionally, temperature monitoring systems for cold-chain logistics employ threshold-based policies. If the temperature exceeds a specified limit (e.g., $8\,^{\circ}\mathrm{C}$), an alert is triggered. While threshold-based systems are easy to implement, they have several major shortcomings. First, there is an excessive number of false alerts: short-term sensor fluctuations, opening/closing the door of a temperature-controlled chamber during transportation, and other common occurrences that have nothing to do with actual temperature mishaps trigger alerts anyway, annoying logistics personnel who have to deal with each alert and spend time determining its source. Second, threshold-based systems do not account for silent failures, i.e., gradual temperature increases or spatial variations in temperature on a truck or ship that do not reach the threshold level but still have the potential to damage pharmaceuticals. Such silent failures are especially insidious since they are completely unpredictable (no alert is triggered) and, consequently, difficult to detect: the only way to determine if a pharmaceutical has been subjected to unsafe temperatures is to test it once it reaches its destination.

The rise of IoT-enabled temperature-monitoring sensors in cold-chain logistics has led to an explosion of data about the temperature, humidity, light, door openings, vibrations, and location of pharmaceutical shipments. These large-scale datasets record readings every 30 seconds on average, generating thousands of data points per shipment. This high resolution enables an unprecedented level of detail in analyzing cold-chain logistics and detecting temperature-related anomalies. However, such a task is not trivial: detecting silent failures requires using machine learning (ML) and deep learning (DL) algorithms to extract meaningful patterns from this high-dimensional data. At the same time, the very nature of ML/DL implies that these algorithms are "black boxes," difficult to interpret. In other words, if an ML algorithm flags a specific shipment as potentially compromised, it is often unclear what the reason for this assessment is. This lack of transparency

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is especially problematic in the current highly regulated pharmaceutical environment, where QA/QC personnel need explicit justification for any deviations before taking action. One potential solution to this issue is provided by the so-called explainable AI (XAI); more specifically, post hoc explainers such as LIME [3] and SHAP [4] enable one to analyze the decisions made by complex ML/DL models and effectively explain them in terms of input features. Therefore, it is reasonable to suggest that in order to design an effective ML-based cold-chain logistics monitoring system, one should prioritize explainable models such as those provided by XAI frameworks rather than rely on traditional threshold-based systems.

The work described in this paper is part of a larger body of research on cold-chain logistics and pharmaceutical temperature monitoring. Specifically, this research makes use of the body of work focused on analyzing temperature-sensor data using ML algorithms. In particular, the paper by Xie et al. [5] describes an innovative approach to cold-chain temperature monitoring: instead of using hard-coded thresholds or simple linear regression to predict temperature changes, Xie et al. construct a sliding-window correlation coefficient-based model that was able to predict temperature deviations with higher precision than previous approaches. This study aims to build upon the work of Xie et al. by focusing on a different (but related) task: developing an ML-based framework for detecting silent failures during cold-chain logistics with an emphasis on model explainability in order to facilitate adoption by QA/QC departments in the pharmaceutical industry.

As a first step towards this goal, this study analyzes the Cold Chain Shipment Silent Failure dataset [6] from Kaggle, a dataset that contains temperature sensor data for 8,000 pharmaceutical shipments along with numerous additional features. The analysis touches upon various aspects of the data, such as the prevalence of different classes (if the shipment was a silent failure or not), missing values, feature types, value ranges, and inter-feature relationships. All of this information is critical for understanding the characteristics of the sample data, which guides the subsequent work of feature engineering, creating predictive models, and analyzing the results with XAI tools.

II. Proposed Methodology

A. Preprocessing

The preprocessing will transform raw shipment-level sensor exports into an analysis-ready dataset to facilitate downstream validation steps. Given the data's nature—a combination of continuous sensor readings, discrete operational counts, and categorical routing information—the features are processed sequentially in the pipeline. First, individual classes of features are validated, exploratory data analysis (EDA) is carried out, and finally, the features are grouped and transformed into engineered variables for modeling. Notably, the imbalance of the target class will be taken into account in all steps, as it is known to have a strong effect on methods for dealing with missing values, outliers, and skewed distributions.

1) Data Inspection and Type Validation: The initial inspection did not reveal any discrepancies in the data structure: the dataset consists of 8,000 records and 24 columns, including a unique shipment identifier (`shipment_id`), 16 continuous (real-valued) features, 6 integer-scaled categorical features, and the target variable `silent_failure`. All columns hold the expected data types, with no coercion attempts observed; therefore, further type conversion will be disallowed to prevent information loss. The `shipment_id` field is unique for all 8,000 records and will be dropped in favor of other features during statistical analysis.

2) Missing Values: There are 64 and 334 missing values in the `temp_recovery_rate` and `rh_max` columns, respectively, with no other fields having missing entries. Since the ratio of missing values vastly exceeds the 5% threshold, no records will be deleted for being incomplete. However, due to the target's dependence on `rh_max`, a naive replacement of NaNs with the column's median will reduce the predictive power of the feature. Therefore, class-conditional median replacement or regression-based NaNs interpolation should be used instead of a simple column-wise substitution.

3) Duplicates and Consistency: There are no identical records in the dataset, and all shipment identifiers are unique. The inspection of ranges revealed that the values of `fill_ratio` and humidity-related features are all within expected ranges. Thus, no obvious entries with physically impossible values have been found in the dataset.

4) Univariate Analysis: The univariate analysis focuses on describing the distributions of real-valued features in the dataset. Firstly, the plots show that there are no strictly normally distributed variables (all have skewness close to zero). Secondly, most of the operational counts (`sensor_gap_hours`, `transit_days`, `door_opens`) follow a Poisson-type distribution, with the mode significantly shifted toward the larger values. Finally, several temperature-related variables show a clear right skew. These patterns are explained in detail in the analysis section.

5) Bivariate Analysis: A comparison of means of all variables given the target class (silent failure) revealed that, on average, the failed shipment method has a higher duration, number of door openings, temperature peaks and valleys, and overall temperature range. Other features do not show a significant difference between the classes. In particular, the boxplots confirm that the interquartile ranges (IQRs) of most variables are similar, but the tails of the distributions are shifted in favor of the failed shipments.

6) Multivariate Analysis, Collinearity, and Redundancy: To find collinear variables, Pearson's correlation matrix was computed across all numeric features. It can be seen that the four temperature-related features, as well as the three humidity-related features, are highly correlated. Additionally, the newly engineered variable `temp_range_c` is very similar to `temp_std_c`, with only a slightly lower variance. Thus, in the case of linear regression, any of the temperature features and only three humidity features (instead of four) should be used to account for multi-collinearity. The three classifying attributes (`carrier_id`, `origin_zone`, `dest_zone`) have negligible correlation with other variables.

7) Outliers: Outliers were identified using the IQR method and marked as entries with values lower than ($Q1 - 1.5 \times \text{IQR}$) and higher than ($Q3 + 1.5 \times \text{IQR}$). The number of outliers varies greatly between variables: from 0.69% for `door_opens` and 0.84% for `vibration_index` to 9.36% for `transit_days`. Since many outliers are linked to the target class, they will be kept in the training set, as records with extreme values are more likely to be involved in a silent failure. Therefore, the next stage's outlier processing should use winsorization for linear models and exclude outliers only when training linear regression.

8) Proposed System Architecture: This EDA stage fits into the system architecture outlined in Figure 1, showing a potential end-to-end pipeline from raw IoT telemetry data to an explanation module for QA and logistics managers.

B. Feature Engineering

Feature engineering aims at transforming raw features into new variables that better capture the process's properties relevant to the prediction task (in this case, silent failures). Specifically, instead of using the mean temperature during transit, which has limited ability to differentiate between records, a range over the transit period can be used as a predictor. Other variables that have been transformed in a similar manner include mean and max humidity, door openings, and vibration intensity. Moreover, `sensor_gap_hours` and max temperature can be combined into one feature, as prolonged periods without readings are likely to be associated with longer temperature excursions. Finally, the analysis shows that the target class is positively correlated with the temperature range and humidity range, negatively correlated with the vibration rate, and weakly correlated with temperature and humidity-related event counts. The correlation coefficients for the engineered variables along with the raw features are detailed in Table III below.

Not all variables benefit from such transformation: for instance, the door-opening rate has a much lower predictive power than the simple count of openings. This is due to the fact that the number of doors opened can be large on short trips, which could bias the model. Therefore, using only the count of doors opened (instead of using it in combination with transit days) is actually preferred in this case. This finding demonstrates in part how new variables are chosen in this pipeline—they are not simply created out of prior knowledge, but rather tested against evidence in the data to see whether they improve predictive performance.

III. Experimental Results

A. Dataset Description

The Cold Chain Shipment Silent Failure Dataset contains 8,000 records and 24 variables, including the binary target `silent_failure` (0/1). Table I contains an overview of the main groups of features used in the study, while Table II reports descriptive statistics of the core numerical features. The target is highly imbalanced as only 19.7% of the records are labeled as silent failures (1), with 6,426 non-failure (0) and 1,574 failure shipments. Consequently, a classifier's performance should be evaluated using its ability to recover positive instances, such as recall, precision, and F1-score, rather than accuracy, which would be uninformative given the strong class imbalance. There are two columns with missing data: `temp_recovery_rate` and `rh_max` contain 64 (0.8%) and 334 (4.18%) missing values, respectively. No duplicate records or shipment IDs were found.

B. Exploratory Data Analysis (EDA)

- **Target class distribution:** 80.3% of all shipments in the Cold Chain dataset do not involve any silent failure (0), while 19.7% are associated with a silent failure (1). The class imbalance is significant but not overwhelming, meaning that predictive modeling should implement stratified sampling and/or class-weighted training to prevent the classifier from always predicting the majority class (Figure 2).
- **Missing values:** Overall, the percentage of missing values is low ($< 5\%$). However, when building a predictive model, special consideration should be given to the `rh_max` column, which has the highest percentage of missing values (Figure 3). If a correlation with other variables is found, the column could be imputed instead of being discarded, as the variable appears to be informative.
- **Univariate distribution:** Variables like `transit_days`, `sensor_gap_hours`, and `temp_std_c` demonstrate a right-skewed distribution, while others like `temp_mean_c`, `rh_mean`, and `door_opens` have an approximately normal distribution with a slight right skew (Figure 4).
- **Bivariate analysis by target:** For all core variables, the median and spread of the distribution for the failure class (1) are higher than for the non-failure class (0) (Figure 5). Specifically, failure shipments have, on average, 6.7 transit days vs. 3.34 days for non-failure, 13.72 door opens vs. 7.6, 5.45 °C temperature vs. 4.01 °C, and a 3.73 °C temperature range vs. 1.99 °C. Across all four variables, higher values are indicative of failure, suggesting that longer, busier shipments with more temperature variation are more likely to experience a silent failure.
- **Correlation matrix:** The temperature variables form a tight group, as do the humidity variables (Figure 6). Notably, `temp_recovery_rate` is correlated with other temperature variables, suggesting that it captures some information about the temperature dynamics. The two sets of weather variables (temperature and humidity) are weakly correlated with one another. (See Table III and Figure 7).
- **Categorical feature analysis:** Transshipment (two-legged) shipments appear to have a higher rate of silent failures compared to single-legged shipments (24.45% vs.

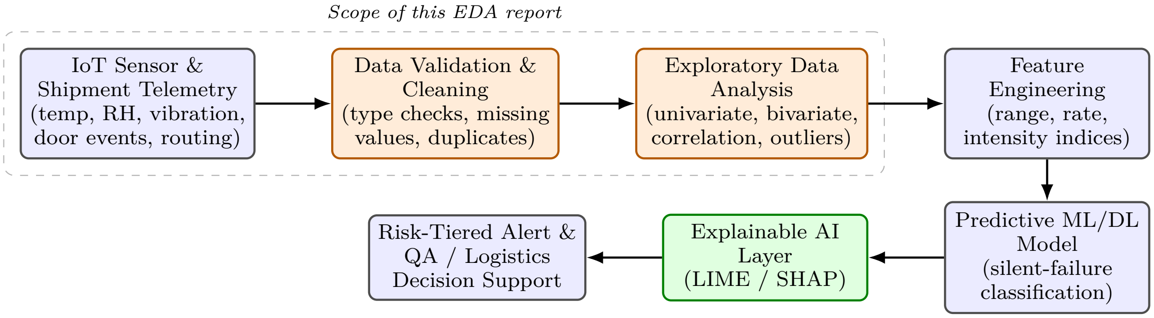

Fig. 1. Proposed end-to-end architecture: from raw cold-chain sensor telemetry, through preprocessing and exploratory data analysis (the scope of this report), to feature engineering, predictive modeling, and an explainable, decision-support output.

TABLE I
Dataset feature groups

Group	**Variables**
Identifier	`shipment_id`
Transit / handling	`transit_days`, `door_opens`, `leg_count`
Temperature	`temp_mean_c`, `temp_max_c`, `temp_min_c`, `temp_std_c`, `temp_recovery_rate`
Humidity	`rh_mean`, `rh_std`, `rh_max`
Packaging / product	`package_type`, `product_volume_l`, `fill_ratio`
Routing	`carrier_id`, `origin_zone`, `dest_zone`
Sensor / motion	`sensor_gap_hours`, `vibration_index`
Anonymized covariates	`feature_x1`, `feature_x2`, `feature_x3`
Target	`silent_failure` (binary)

TABLE II
Descriptive statistics for core numerical features

Variable	**Mean**	**Std**	**Min**	**25%**	**50%**	**75%**	**Max**
transit_days	4.00	2.96	0.50	1.82	3.14	4.78	17.33
door_opens	8.80	6.17	0.00	4.00	7.00	13.00	36.00
temp_mean_c	4.29	1.72	-1.67	3.23	4.12	5.21	12.15
temp_max_c	5.82	2.46	-1.38	4.18	5.28	7.11	19.38
temp_min_c	3.49	1.73	-4.33	2.43	3.48	4.41	12.11
temp_std_c	1.27	0.70	0.00	0.71	1.12	1.72	4.22
temp_recovery_rate	0.44	0.14	0.01	0.35	0.44	0.54	0.89
rh_mean	0.51	0.10	0.22	0.44	0.50	0.57	0.91
rh_std	0.07	0.04	0.00	0.04	0.06	0.09	0.29
rh_max	0.62	0.15	0.24	0.51	0.59	0.71	0.99
product_volume_l	12.03	3.93	0.08	9.40	12.04	14.69	29.28
fill_ratio	0.72	0.14	0.19	0.62	0.72	0.82	1.00
sensor_gap_hours	0.79	0.79	0.00	0.22	0.55	1.09	6.00
vibration_index	1.83	0.93	0.00	1.17	1.78	2.42	5.82

14.64%) (Figure 8). Similarly, package type 1 has a higher rate compared to package type 0 (23.56% vs. 17.51%). In contrast, there is little variation in the rate across levels of the `carrier_id` (range between 16.7% and 22.6%) and `origin_zone`/`dest_zone` (range between 18.2% and 21.3%). The anonymized `feature_x3` is nearly identical for the two observed values (19.9% vs. 19.1%).

- **Engineered feature analysis:** The density plot for the temperature range is shifted towards higher values for the failure class compared to the non-failure class (Figure 9). Similarly, the density plot for the vibration rate is shifted towards lower values for the failure class, which is consistent with the negative correlation between the vibration rate and target found earlier.
- **Transit duration and peak temperature:** Higher peak temperatures are associated with longer shipment duration (Figure 10). Silent failures tend to be associated with higher values of both variables, while non-failure shipments are concentrated in lower values.

Taken together, the findings from this EDA support the initial hypothesis that silent failures in this dataset are caused by complex, interacting factors rather than a single characteristic. Specifically, longer shipment duration, additional shipment legs, and greater variation in temperature appear to contribute to silent failures, while other variables, such as carrier, origin/destination zone, and package type, have limited predictive power. This conclusion provides a theoretical foundation for the

Fig. 2. Target class distribution: silent-failure shipments form the minority class (19.7%).

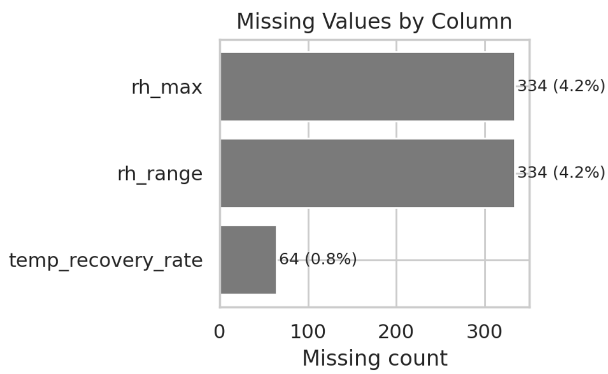

Fig. 3. Missing values by column: only `temp_recovery_rate` and `rh_max` are affected.

TABLE III
FEATURE CORRELATION WITH `SILENT_FAILURE` (TOP MAGNITUDE, SELECTED)

Feature	**Pearson r**
transit_days	0.453
temp_max_c	0.441
door_opens	0.395
temp_std_c	0.392
temp_range_c (eng.)	0.387
rh_std	0.380
temp_mean_c	0.333
rh_max	0.324
rh_mean	0.319
vibration_rate (eng.)	-0.263
excursion_intensity (eng.)	0.197
rh_range (eng.)	0.176

predictive modeling component of this project.

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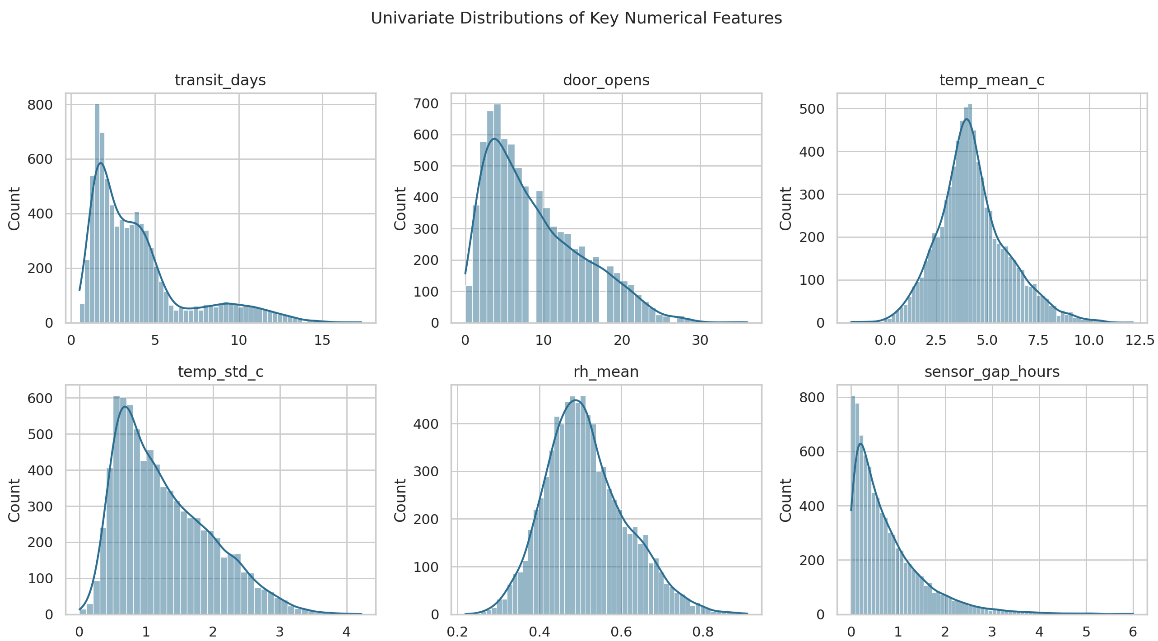

Fig. 4. Univariate distributions of key numerical features.

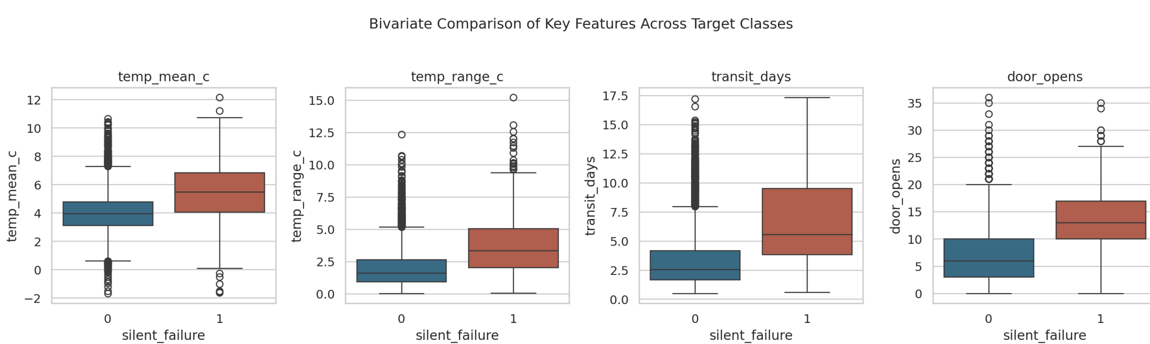

Fig. 5. Bivariate comparison of key features across target classes (0 = non-failure, 1 = silent failure).

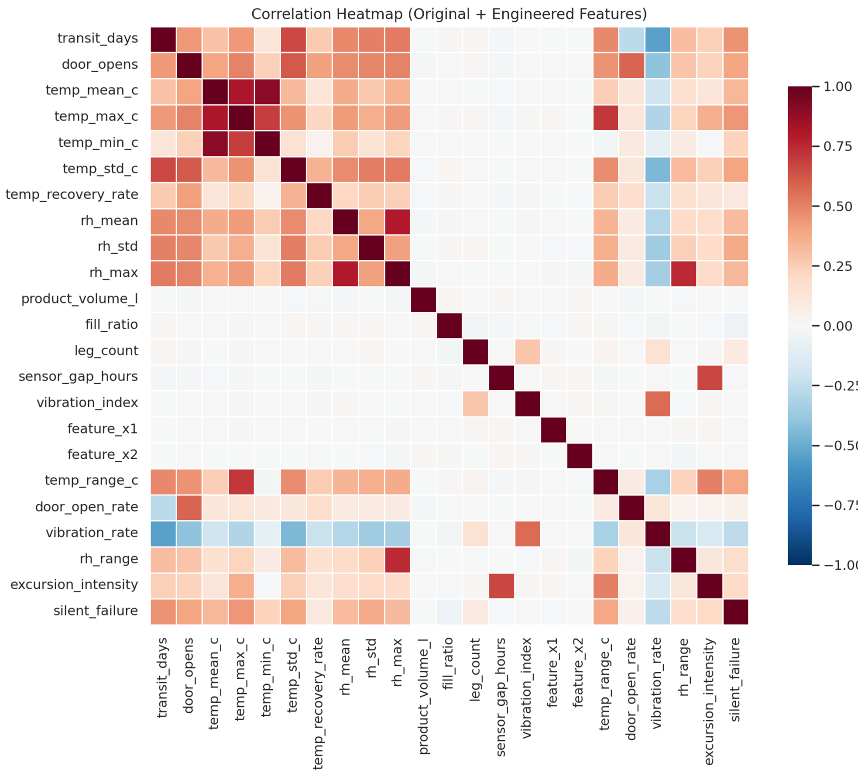

Fig. 6. Correlation heatmap across original and engineered numerical features.

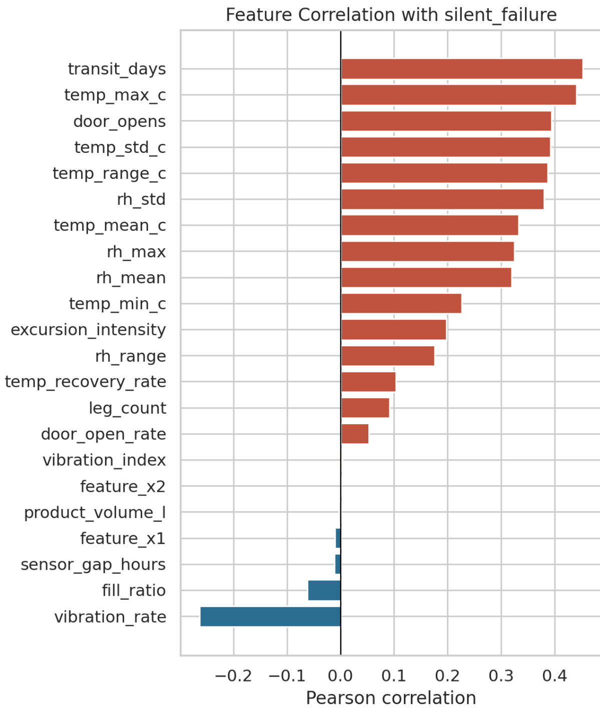

Fig. 7. Ranked feature correlation with `silent_failure`.

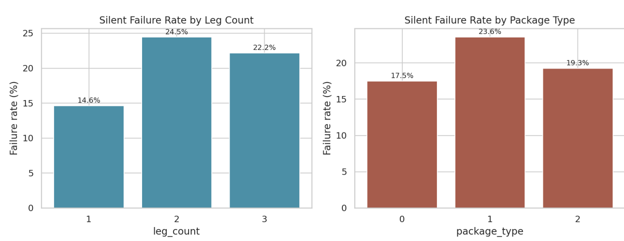

Fig. 8. Silent-failure rate by leg count and package type.

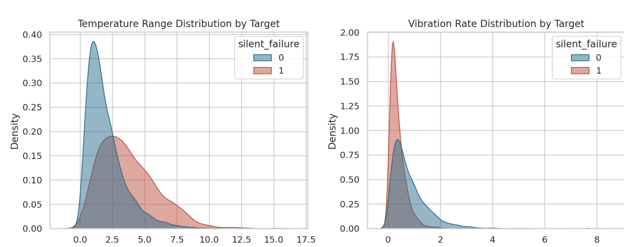

Fig. 9. Class-conditional distributions of two engineered features: temperature range and vibration rate.

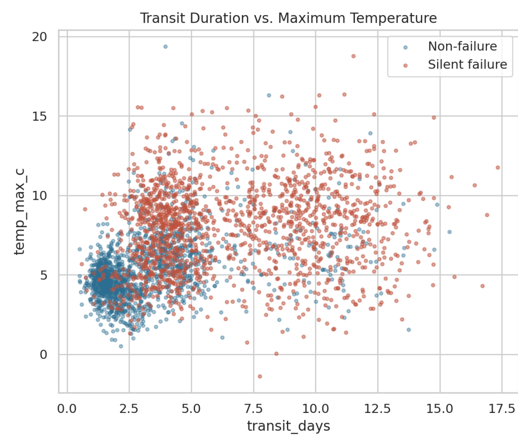

Fig. 10. Transit duration vs. maximum temperature, colored by target class.